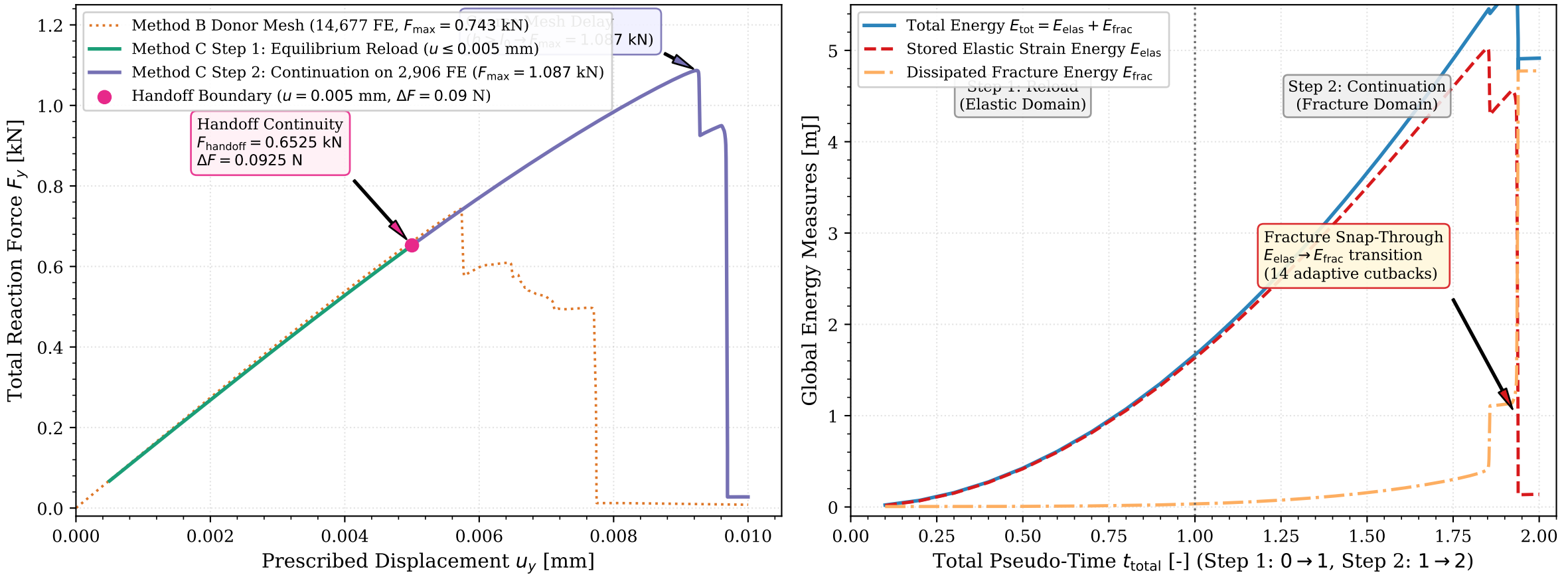
(a) Load-Displacement Response & Handoff Continuity
Method B Donor Mesh (14,677 FE, $F_{max} = 0.743$ kN)
Method C Step 1: Equilibrium Reload ($u \leq 0.005$ mm)
Method C Step 2: Continuation on 2,906 FE ($F_{max} = 1.087$ kN)
Handoff Boundary ($u = 0.005$ mm, $\Delta F = 0.09$ N)
Handoff Continuity
$F_{handoff} = 0.6525$ kN
$\Delta F = 0.0925$ N
Total Reaction Force F_y [kN]
Prescribed Displacement u_y [mm]
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.000
0.002
0.004
0.006
0.008
0.010
(b) Global UEL Energetic Partition & Dissipation
Total Energy $E_{tot} = E_{elas} + E_{frac}$
Stored Elastic Strain Energy E_{elas}
Dissipated Fracture Energy E_{frac}
Step 2: Continuation
(Fracture Domain)
Fracture Snap-Through
$E_{elas} \rightarrow E_{frac}$ transition
(14 adaptive cutbacks)
Global Energy Measures [mJ]
Total Pseudo-Time t_{total} [-] (Step 1: 0 → 1, Step 2: 1 → 2)
0
1
2
3
4
5
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00